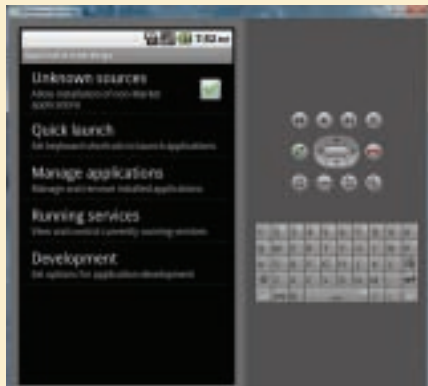


download and click the respective download buttons for those apps.

Android apps exist in files that have the APK (Android Package) file extension. An



Settings for installing non-market apps

APK file is actually a package that contains a number of bundled components—similar to how a zip file works. In fact, when you download an Android app file, it might have either an APK or ZIP file extension. If you download an Android app that has a ZIP extension, don't try to unzip it—just change the ZIP extension to APK.

So how do you install an Android app that you've downloaded onto your

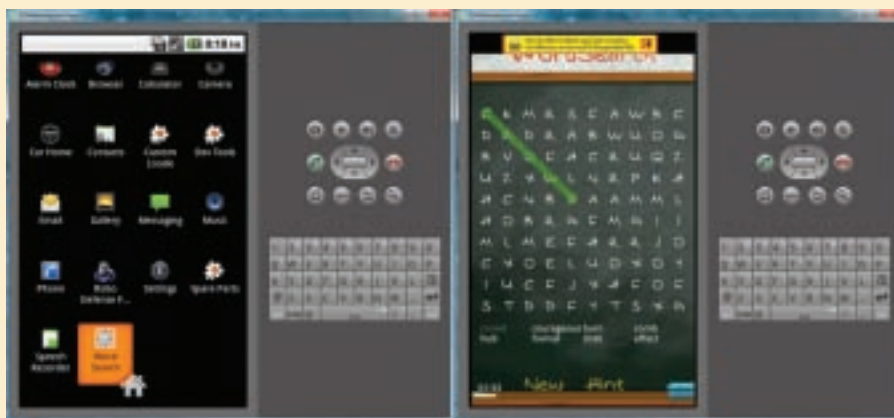
computer's hard drive? The first thing you need to do is fire up the AVD and wait for it to launch. Once the AVD is running, make sure that it's set to permit the installation of non-Market applications. Do this by selecting the Launcher icon, choose the Settings icon, click on Applications, and make sure that Unknown sources is enabled.

With the AVD running, open a Windows Command-Prompt window (or what we old-schoolers like to call a DOS-Prompt window). You'll be typing a command that executes the adb.exe application that lives in the android-sdk-windows/tools folder, followed by a space, then the word install, another space, and then the Android app's file name. The trick is, you also need to include the full paths for both the adb.exe app and the APK file you are installing (unless they both live in the same folder).

Here's an example of how we installed the Word Search app in my AVD:

```
c:\android-sdk-windows\tools\adb.exe install c:\temp\wordsearch.apk
```

It might take a few moments to install. When the word "Success" appears in the Command Prompt window, you know it has installed. I've found that on rare occasions, installation will fail on the first try, but succeed on a second attempt.



Once you've entered the Settings, you can download apps from external sources



Uninstalling an app is also simple

Now if you go back to the AVD, you should see an icon for the app you just installed appear amongst the other apps. Launch the app and it should run as expected.

If you want to uninstall an app, do so by going to Settings > Applications > Manage Applications > Downloaded, then select the application you want to delete, and click the Uninstall button.

If the AVD actually had access to the Android Market, then there would be a bevy of Uninstall applications

to choose from—which would make uninstalling apps a much easier task. We didn't have much luck finding a working uninstall app from the independent Android apps stores.

Accessing the Android Market in an AVD

If you want to be able to access the Android Market in an AVD,

you can do it; but it will need to be with an older version of the Android OS. A software Engineer by the name of Kumar Bibek posted instructions on his blog; on how to do this using a file downloaded from the HTC Developer Center Website with an Android 1.6 AVD. While some users claimed it worked for them, we couldn't get it to work. But luckily, Mr. Bibek made an alternative custom file available for download that worked for me. Here's what you need to do:

1. Create an AVD that uses Android 1.6—but don't start it yet.
2. Download the system.img file that Mr. Bibek made available here.
3. Copy the system.img file you just downloaded to the AVD folder for the AVD you just created. You can find this folder in ...\.android\avd in your user folder. For example, we created an AVD called "Market" and this is the path to its folder on my system: C:\Users\ExtremeTech\.android\avd\Market.avd

4. Start the AVD for the first time; but when the Launch Options window opens, enable the Wipe user data option. Then click the Launch button. You should only have to do this once—on subsequent launches of the AVD, disable the Wipe user data option.

5. Once the AVD loads, you'll have to go through a few setup screens; but you can skip most of them.

6. Navigate over to the Applications page and you should see an icon for the Android Market, as well as a few other apps that weren't present in the Android 2.2 AVD—such as Google Maps and Google Talk.

7. Before you can access the Android Market you'll have to either create a Google account or enter the login credentials for an existing Google account.

8. Once your Google account credentials are entered, you can access the Android Market and start downloading apps into the AVD!

Have fun exploring Android!